Antenna: Resilient Collective Robotics for Rice Paddies

The Kinematic Quintet | **Track:** Embodied Robotics & Autonomous Agents

Codebase: https://github.com/als0fia/AgentVerse_Kinematic_Quintet | **Target:** NVIDIA Jetson Orin Nano (10W Edge)

1 Abstract: Overview & Purpose

Rice feeds over half the world, yet planting in muddy fields still relies on an aging workforce—harming their health, threatening sustainability, and slowing harvests. **Antenna** is a team of small robots (**"Ennas"**) that tests the soil, plants missing rice to maximize yield, and smartly recovers stuck robot units to maximize farm output.

2 Why It Matters: Socio-Economic Need

- **Labor & Mechanization Deficit:** Aging farmers and rising wages (£22–£40/day) threaten rushed planting windows ($<$ 10 days). While national tillage exceeds 90%, rice planting mechanization is only **67.6%** (MARA 2024/2026; CSAE 2025), with severe shortages in Southern mountainous regions with small, fragmented fields ($<$ 0.15 ha).
- **Why Current Machines Fail & Why Antenna Wins:** Heavy transplanters (700–1,000 kg) sink in 15–50 cm mud pans, requiring 4–6 workers to pull out while crushing crops. Antenna provides affordable robotic farming services to China's ≈80,600 agricultural cooperatives at **£6.50–£10 per mu** (vs. £17–£25 for manual labor), supported by 30–35% government hilly-machinery subsidies.

Table 1: **Efficiency and Cost Benefits of Agricultural Automation**

Technology	Efficiency Gain	Cost Reduction
Manual Rice Farming	Baseline	Baseline
Drone Seeding & Monitoring	40%	20%
Robotic Rice Planters	42%	30%
AI-Powered Irrigation	50%	40%
Automated Plant Growing	60%	35%

UN Sustainable Development Goals met: 2, 11, 12, 13.

3 The Robot Hardware: Lightweight "Enna" Design

Every swarm unit shares an identical **"Enna"** architecture: onboard edge computing (Jetson Orin Nano), soil-sensing and planting tools, and a rigid rescue hitch:

- **Compact Low-Pressure Hull:** A lightweight (**4.2 kg**), waterproof body on gripped paddle wheels that floats over soft mud without sinking, built narrow enough (**276 mm** wide) to drive cleanly inside standard **300 mm** rice rows without crushing plants.
- **Onboard Tools:** Equipped with a plunging soil probe to test mud quality, a gravity-fed seed dropper, **3-beam** forward laser distance sensors, and a front lifting hitch that locks onto stuck teammates to pull them free.

4 The Hardware: Sensors & Actuators

- **Sensors (Perceive):** (1) **3-Beam LiDAR** (front obstacles & terrace walls); (2) **Whiskers** (crop stem touch for row centering); (3) **RTK + Odometry** (position & wheel-slip stall detection); (4) **Cone Penetrometer** (mud firmness, moisture, & missing plant gaps); (5) **P2P Radio** (local SOS & map sharing within 2.15 m).
- **Actuators (Act):** (1) **4-Wheel Drive** (skid-steer motion at $\pm$0.45 m/s); (2) **Farming Tools** (seed-drop valve & underbelly weeders); (3) **Rescue Tools** (front lifting boom to break mud suction, rotary cam-latch lock, & rear anchor spades).

5 Perceive: Soil Checks & Stall Detection

Ennas rely strictly on their own onboard sensors:

1. **Checking Soil Before Planting:** Each robot probes soil moisture (0.25–0.80), firmness (0.20–0.90), and empty plant spacing ($\geq$ 0.60). Validated planting spots are shared directly with nearby teammates over local radio.
2. **Detecting When Stuck in Mud:** If a robot commands its wheels to drive ($\geq$ 0.04 m/s) with no obstacle ahead ($>$ 0.32 m), yet barely moves forward ($<$ 0.005 m) or turns ($<$ 0.02 rad), it flags that it is bogged down in mud.

6 Decide: AI Agent

Our system pairs a Small Language Model with physics learning to recover trapped Ennas:

- **SLM Strategist + Residual Neural Learner:** An on-edge Small Language Model (**Qwen 2.5 0.5B**) acts as the high-level strategist, querying an online **Residual Neural Learner** to calculate real-world mud friction and wheel sinkage before deciding to self-push or call a teammate for a tow.
- **Modal GPU Training & Edge Efficiency:** Fine-tuned on **Modal's cloud GPUs** in under five minutes, achieving **97.88% accuracy** on unseen test scenarios within a **10-Watt edge power budget**.
- **Hardcoded Safety Gate & 2 ms Fallback:** Every AI decision passes through a **hardcoded safety gate** that blocks unsafe commands and protects battery limits, backed by an instant **2-millisecond automatic fallback**.

7 Act: Smart Swarm Rescue

When a robot gets trapped in soft mud ($<$ 30 kPa), the mud's suction makes it difficult to escape. Sending another robot without assessing the conditions could trap both, causing a **critical failure** known as cascade entrapment.

- **Detecting and Communicating the Problem:** The trapped robot detects its lack of movement, stops its wheels, estimates the mud resistance, and sends an SOS through local P2P radio to nearby teammates without needing internet.
- **Evaluating Rescue Risks:** The swarm calculates whether rescuing the robot is worth the risk, considering its remaining seeds, the time lost from planting, and the danger of another robot getting stuck:

$$\text{Net} = \text{Rescue Benefit} - \text{Opportunity Cost} - \text{Hazard Risk}$$

- **Making a Safe Decision:** If Net $>$ 0, the nearest available robot attempts a rescue while the others continue planting. If Net $\leq$ 0, the swarm refuses rescue to prevent further robots from getting trapped, prioritising the remaining mission.

8 Physical Rescue: How One Robot Pulls Out Another

- **Tandem Docking & Suction Break:** The rescuer aligns into the stuck robot's rear V-funnel and snaps its rotary cam-latch shut to lock both robots into a rigid 8-wheel train. It then tilts its front boom upward ($+Z$) to pry the stuck robot up and break mud suction while dropping rear anchor spades before pulling backward.
- **MuJoCo Physics Validation:** Verified $>$ 0.08 m extraction pull within safe force limits ($<$ 250 N, tilt $<$ 0.12 rad), followed by immediate mission resumption.

9 Control Speeds & System Timing

Control Layer	Agent / Swarm Responsibility	Hardware	Speed
Tier-0 Safety	Hardcoded safety gate & auto fallback	MCU/CPU	**2** ms
Tier-1 Swarm	Physics learner, P2P radio, task auction	CPU	10–15 ms
Tier-2 AI Brain	Qwen 2.5 0.5B high-level strategist	Jetson (10W)	$<$ 120 ms

10 Operating Limits & Real-World Scope

- **Mud Depth Limit:** Effective in mud up to $\leq$ 45 cm deep. Deeper liquid slurry ($>$ 60 cm) exceeds 8-wheel traction, triggering safe refusal for human pickup.
- **Simulation vs. Field Trials:** Multi-body docking and P2P radio are validated in 3D MuJoCo; wet clay slurry and field trials with co-ops are scheduled for Phase-2.
- **Target Farm Size:** Designed for gap-replanting in small hilly terraces ($<$ 0.15 ha), complementing rather than replacing large tractors on flat plains.

11 Safety & Operational Resilience

- **Operator UI Emergency Stop:** The mission dashboard features a continuous, real-time manual E-Stop that instantly overrides AI policies, cuts motor torque, and locks all chassis brakes in $<$ 2 ms.
- **Environmental Ingress Protection (IP68):** Sealed waterproof hull with a positive-pressure air cavity protects internal electronics, LiDAR wiring, and batteries across saturated 15–50 cm flooded paddy mud pans (*Lannitian*), topped by an elevated splash-proof BeiDou RTK dome.
- **Zero Prompt Injection (Air-Gapped Edge AI):** All models (Qwen 2.5 0.5B and Residual Learner) execute fully on-device on the 10W Jetson chip with zero remote API dependencies or external text interfaces, completely eliminating cyber prompt-injection and cloud-hijacking attack surfaces.
- **Decentralized Mesh (No Single Point of Failure):** Fully peer-to-peer (P2P) mycelial gossip over 2.15 m ad-hoc links eliminates central server bottlenecks. If an Enna gets stranded or loses power, surviving peers reallocate tasks autonomously without fleet-wide disruption.